```
-!pip install gensim
```

```
Collecting gensim
  Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (8.4 kB)
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/dist-packages (from gensim) (2.0.2)
Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from gensim) (1.16.3)
Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim) (7.5.0)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1->gensim) (2.0.1)
Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (27.9 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 27.9/27.9 MB 25.5 MB/s eta 0:00:00
Installing collected packages: gensim
Successfully installed gensim-4.4.0
```

```
import numpy as np
from gensim.models import KeyedVectors
```

```
embedding_path = "/content/GoogleNews-vectors-negative300.bin.gz"

print("Loading Google News Embeddings...")
model = KeyedVectors.load_word22vec_format(embedding_path, binary=True)
print("Model Loaded Successfully!")
```

```
Loading Google News Embeddings...
Model Loaded Successfully!
```

```
def association(w, A, B, model):
    return np.mean([model.similarity(w, a) for a in A]) - np.mean([model.similarity(w, b) for b in B])

def weat_score(X, Y, A, B, model):
    return np.sum([association(x, A, B, model) for x in X]) - np.sum([association(y, A, B, model) for y in Y])

def effect_size(X, Y, A, B, model):
    scores = [association(w, A, B, model) for w in X + Y]
    mean_diff = np.mean([association(x, A, B, model) for x in X]) - np.mean([association(y, A, B, model) for y in Y])
    return mean_diff / np.std(scores)
```

```
male_words = ["man", "male", "boy", "brother", "he", "him", "son"]
female_words = ["woman", "female", "girl", "sister", "she", "her", "daughter"]

career_words = [
    "executive", "management", "professional",
    "corporation", "salary", "office", "career"
]

family_words = [
    "home", "parents", "children", "family",
    "cousins", "marriage", "wedding"
]
```

```
print("Computing WEAT Gender-Career Bias...")

gender_bias = effect_size(
    male_words,
    female_words,
    career_words,
    family_words,
    model
)

print("\n-------------------- RESULT --------------------")
print(f"WEAT Effect Size : {gender_bias:.4f}")

if gender_bias > 0:
    print("Bias Direction: Career is more associated with MALE terms")
elif gender_bias < 0:
    print("Bias Direction: Career is more associated with FEMALE terms")
else:
    print("No detectable bias")

print("------------------------------------------------")
```

```
Computing WEAT Gender-Career Bias...
```

```
-------------------- RESULT --------------------
WEAT Effect Size : 0.3136
Bias Direction: Career is more associated with MALE terms
------------------------------------------------
```

```
Start coding or generate with AI.
```